Emissions Compared with Benchmarks

> Part of the Functional Series. This document compares the platform's emissions series with the Global Carbon Budget, the UNEP Emissions Gap Report and the IPCC. The scope columns matter more than the numbers: the sources agree closely once their definitions are aligned, and disagree by several gigatonnes when they are not.

1. Fossil CO_2 in 2024

Source	2024 fossil CO_2 ($GtCO_2$)	Gases	Bunkers (international aviation and shipping)	Cement carbonation sink
Platform, OWID World row	38.6	CO_2	Included (1.19)	Not netted out, as far as can be checked
Platform, PRIMAP-hist	37.97	CO_2	Excluded	n/a
Global Carbon Budget 2025	37.8 ± 1.8	CO_2	Estimated separately; whether the headline total includes them was not confirmed in the text read	**Netted out** (0.2 GtC ≈ 0.7 $GtCO_2$)
UNEP EGR 2025 (EDGAR-based)	39.6 ± 3.2	CO_2	Included	Not stated

The gap between the platform's 38.6 and the Global Carbon Budget's 37.8 is about 0.8 $GtCO_2$. Subtracting the cement carbonation sink the Global Carbon Budget nets out (0.2 GtC, about 0.73 $GtCO_2$) leaves 37.9, within rounding of 37.8. The platform's number therefore appears to be the gross figure, but OWID's codebook does not state its treatment, so this is a consistent explanation, not a verified one. Either way the two agree within the Global Carbon Budget's own uncertainty of ±1.8.

The same alignment works for the totals including land use:

	Fossil CO_2	Land-use CO_2	Total anthropogenic CO_2
Platform (OWID / Global Carbon Project), 2024	38.6	4.6	43.2
Platform, less the 0.73 carbonation sink	37.9	4.6	42.4
Global Carbon Budget 2025, 2024	37.8 ± 1.8	about 4.8 (1.3 ± 0.7 GtC)	42.4 ± 3.2

The Global Carbon Budget's 2025 projection (38.1 $GtCO_2$ fossil, up 1.0%, and 4.1 $GtCO_2$ land use) is beyond the end of the platform's data, which stops at 2024.

Land use is the largest relative uncertainty. Its uncertainty is about ±2.6 $GtCO_2$ a year (one standard deviation), compared with ±1.8 for all fossil CO_2.

2. Total greenhouse gases

The totals look very different, and almost all of the difference is scope:

	Platform (PRIMAP-hist) 2024	UNEP EGR 2025, 2024	Difference
Fossil CO_2	37.97	39.6	+1.6
Land-use CO_2 (bookkeeping)	not included	4.4	+4.4
Methane (CO_2e)	8.42	9.3	+0.9
Nitrous oxide (CO_2e)	2.73	2.6	–0.1
F-gases (CO_2e)	1.45	1.7	+0.25
Total ($GtCO_2e$)	**50.55**	**57.7 ± 5.5**	**+7.1**
GWP set	AR5 GWP-100	AR6 GWP-100	

An approximate decomposition of the 7.1 $GtCO_2e$ gap, not a reconciliation:

- **Land-use CO_2 (4.4)** is the biggest single item, a scope difference the platform states.
- **Fossil CO_2 (+1.6)** is mostly the international aviation and shipping that PRIMAP-hist excludes (about 1.2 in OWID's 2024 figures); the remainder reflects different inventories.
- **Methane (+0.9)** is the largest gap that is not a scope difference. It was not traced here. A similar pattern appears against the IPCC's 2019 estimate (below).
- Nitrous oxide and F-gases nearly cancel.

The same check for 2019 against the IPCC. AR6 Working Group III estimates net anthropogenic emissions in 2019 at 59 ± 6.6 $GtCO_2e$, made up of:

Component, 2019	IPCC AR6 WGIII ($GtCO_2e$)	Platform (PRIMAP-hist, OWID)	Note
Fossil CO_2 and industry	38 ± 3	37.1 (OWID, incl. bunkers); 36.4 (PRIMAP-hist, excl.)	Consistent
Land-use CO_2	6.6 ± 4.6	4.9 (OWID / Global Carbon Project)	Within the IPCC's range
Methane	11 ± 3.2	8.2	Platform is lower
Nitrous oxide	2.7 ± 1.6	2.6	Consistent
F-gases	1.4 ± 0.41	1.2	Consistent
Total	**59 ± 6.6**	**48.3** (PRIMAP-hist, no land use)	Platform total excludes land use

Excluding land use, the IPCC's own components sum to about 53.1, against the platform's 48.3, a gap of about 4.8. Methane (2.8) and fossil CO_2 (about 1.6, largely the bunkers PRIMAP-hist excludes) account for most of it.

3. Growth in 2024

Source	2024 growth	What it covers
Platform, OWID fossil CO_2	+1.3%	Fossil CO_2 with bunkers

Source	2024 growth	What it covers
Global Carbon Budget 2025	+1.1%	Fossil CO_2
EDGAR, as quoted by UNEP	+1.1%	Fossil CO_2
Platform, PRIMAP-hist total GHG	+1.0%	CO_2, CH_4, N_2O, F-gases; no land use
UNEP, total GHG	+2.3%	All gases including land use
UNEP, total GHG excluding land use (derived from its Table 2.1)	about +1.3%	All gases without land use

The headline difference (+2.3% against about +1%) is almost entirely land use: UNEP's net land-use CO_2 rose from 3.6 to 4.4 $GtCO_2$, a 21% jump (UNEP's own statement). Without it, the sources agree within a few tenths of a percent. This is the main reason for keeping land use out of trend comparisons.

4. Cumulative emissions

	Platform	IPCC AR6	Comparison
Cumulative net CO_2, 1850–2019	2,542 $GtCO_2$ (fossil 1,658; land use 884)	2,400 ± 240 $GtCO_2$ (Synthesis Report A.1.3)	Within the IPCC's range
Share emitted 1990–2019	41.8%	"about 42%"	Matches

5. What this comparison shows

- **Levels agree once scope is aligned.** Fossil CO_2 matches the Global Carbon Budget within rounding after the cement carbonation sink; land use and cumulative totals sit inside the IPCC and Global Carbon Budget ranges.
- **The platform's total GHG is lower by design, not by error.** It excludes land use and international transport, uses an older GWP set, and (not traced here) carries less methane than the global inventories.
- **Trends agree better than levels.** Growth in 2024 is within a few tenths of a percent across sources once land use is set aside.
- **The 2025 projections are not yet in the platform.** The Global Carbon Budget projects fossil CO_2 of 38.1 $GtCO_2$ for 2025, a record.

See also

- The Reference Landscape
- Temperature and Concentration Compared with Benchmarks
- Data Quality and Harmonization: the known discrepancies between the platform's own sources